

COP2800C Module 3 Graded Programming Assignment

Make the following changes to the Penguin Rookery from Module 2:

- Replace the Species class with an enumerated type.
- Replace the three Penguin instance variables in the rookery with an array of Penguins.
- If you do not already have a default constructor for the PenguinRookery, add one.
- Add an overloaded PenguinRookery constructor which accepts an array of Penguins as a parameter. Name the parameter using the same name as your array instance variable and use the **this** reference to assign to the instance variable in the constructor.
- Add a toString method to the PenguinRookery which displays the state of the rookery, including the penguin states and total number of penguins in the rookery.
- Change the species instance variable in the Penguin class from int to the enumerated type.
- Change any parameters in the Penguin class which use the int type for the species to the enumerated type.
- Add an accessor and mutator to the Penguin species to the Penguin class. Use the **this** reference where needed (required in the mutator, optional in the accessor).
- Change the toString method in the Penguin class to use the name() method of the enumerated type.
- Change the unit tests (main method) in the PenguinRookery class to instantiate the rookery object using the overloaded constructor; pass a locally instantiated array of penguins to it.
- Add a unit test for the Species mutator and accessor in the main method by modifying the Penguin array elements' species and then redisplaying the values (refer to the practice exercise solution for an example of how this should be done).

Expected output is as follows:

```
Our rookery contains the following penguins:  
First penguin is a(n) CHINSTRAP.  
Second penguin is a(n) GENTOO.  
And the final penguin is a ADELIE.  
There are a total of 3 penguin species in our rookery.  
Testing mutators and accessors  
first penguin: ADELIE  
second penguin: CHINSTRAP  
third penguin: GENTOO
```

Include all source code in a single source file (PenguinRookery.java).

Be sure your code is indented properly and includes a 4-line ID header.